

Gibson–Schwartz PDE Derivation

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1 Setup under the physical measure $\mathbb{P}$

Work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$, with spot price S_t and convenience yield δ_t evolving as

$$dS_t = (\mu - \delta_t) S_t dt + \sigma_1 S_t dW_1^\mathbb{P}, \quad (1)$$

$$d\delta_t = \kappa(\alpha^\mathbb{P} - \delta_t) dt + \sigma_2 dW_2^\mathbb{P}, \quad (2)$$

with correlated drivers, $dW_1^\mathbb{P} dW_2^\mathbb{P} = \rho dt$, $\rho \in (-1, 1)$.

2 Change to the risk-neutral measure $\mathbb{Q}$

Under $\mathbb{Q}$ the traded asset must drift at the risk-free rate r , so we introduce a market price of risk λ_i for each driver. By Girsanov's theorem,

$$dW_i^\mathbb{Q} = dW_i^\mathbb{P} + \lambda_i dt, \quad i = 1, 2, \quad (3)$$

with the measure change defined by the Radon–Nikodým derivative

$$\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{F}_t} = \exp \left(- \int_0^t \lambda_1 dW_1^\mathbb{P} - \int_0^t \lambda_2 dW_2^\mathbb{P} - \frac{1}{2} \int_0^t (\lambda_1^2 + \lambda_2^2) ds \right). \quad (4)$$

2.1 Substituting the measure change

Rearranging Equation 3 gives $dW_i^\mathbb{P} = dW_i^\mathbb{Q} - \lambda_i dt$. Substituting into Equations 1 and 2:

$$dS_t = [(\mu - \delta_t) - \sigma_1 \lambda_1] S_t dt + \sigma_1 S_t dW_1^\mathbb{Q}, \quad (5)$$

$$d\delta_t = [\kappa \alpha^\mathbb{P} - \sigma_2 \lambda_2 - \kappa \delta_t] dt + \sigma_2 dW_2^\mathbb{Q}. \quad (6)$$

These are correct but leave λ_1, λ_2 unspecified, and the convenience-yield drift is no longer in mean-reverting form. We resolve each in turn.

2.2 Pinning down λ_1 : the no-arbitrage drift condition

The economic content of no-arbitrage here is the cost-of-carry condition. In Equation 1, μ is the *total* expected return on holding the physical commodity: the price drift is $\mu - \delta_t$, and the holder additionally collects the convenience yield δ_t as a non-pecuniary dividend

(avoiding stock-outs, retaining operational flexibility), the two summing to μ . Under $\mathbb{Q}$, no-arbitrage requires the total expected return on the traded spot to equal the risk-free rate r . Since the convenience yield accrues to the holder like a dividend, this is equivalent to a risk-neutral *price* drift of $r - \delta_t$. Matching this to the drift in Equation 5,

$$(\mu - \delta_t) - \sigma_1 \lambda_1 = r - \delta_t \quad \implies \quad \lambda_1 = \frac{\mu - r}{\sigma_1}, \quad (7)$$

which identifies λ_1 as the market price of spot risk (the Sharpe ratio of the spot process). Substituting back:

$$dS_t = (r - \delta_t)S_t dt + \sigma_1 S_t dW_1^{\mathbb{Q}}. \quad (8)$$

2.3 Pinning down $\alpha^{\mathbb{Q}}$: restoring mean-reverting form

Factoring κ back out of the drift in Equation 6:

$$\kappa \alpha^{\mathbb{P}} - \sigma_2 \lambda_2 = \kappa \left(\alpha^{\mathbb{P}} - \frac{\sigma_2 \lambda_2}{\kappa} \right) =: \kappa \alpha^{\mathbb{Q}}. \quad (9)$$

Unlike λ_1 , the convenience-yield risk premium λ_2 is not fixed by a no-arbitrage argument, since δ_t is not a traded asset; it is identified jointly with the other parameters by calibration to the observed futures term structure. Substituting back:

$$d\delta_t = \kappa(\alpha^{\mathbb{Q}} - \delta_t) dt + \sigma_2 dW_2^{\mathbb{Q}}, \quad \alpha^{\mathbb{Q}} = \alpha^{\mathbb{P}} - \frac{\sigma_2 \lambda_2}{\kappa}. \quad (10)$$

Only the long-run level shifts ($\alpha^{\mathbb{P}} \rightarrow \alpha^{\mathbb{Q}}$); κ and σ_2 are unchanged.

3 Itô's lemma and the futures PDE

Let $F = F(S_t, \delta_t, t)$ be the futures price. By the bivariate Itô lemma,

$$dF = \frac{\partial F}{\partial t} dt + \frac{\partial F}{\partial S} dS_t + \frac{\partial F}{\partial \delta} d\delta_t + \frac{1}{2} \frac{\partial^2 F}{\partial S^2} (dS_t)^2 + \frac{\partial^2 F}{\partial S \partial \delta} (dS_t)(d\delta_t) + \frac{1}{2} \frac{\partial^2 F}{\partial \delta^2} (d\delta_t)^2. \quad (11)$$

The quadratic-variation terms, with $dW_1^{\mathbb{Q}} dW_2^{\mathbb{Q}} = \rho dt$, are

$$(dS_t)^2 = \sigma_1^2 S_t^2 dt, \quad (d\delta_t)^2 = \sigma_2^2 dt, \quad (dS_t)(d\delta_t) = \rho \sigma_1 \sigma_2 S_t dt. \quad (12)$$

Substituting the $\mathbb{Q}$ -dynamics gives

$$\begin{aligned} dF = & \left[\frac{\partial F}{\partial t} + (r - \delta_t) S_t \frac{\partial F}{\partial S} + \kappa(\alpha^{\mathbb{Q}} - \delta_t) \frac{\partial F}{\partial \delta} + \frac{1}{2} \sigma_1^2 S_t^2 \frac{\partial^2 F}{\partial S^2} \right. \\ & \left. + \rho \sigma_1 \sigma_2 S_t \frac{\partial^2 F}{\partial S \partial \delta} + \frac{1}{2} \sigma_2^2 \frac{\partial^2 F}{\partial \delta^2} \right] dt + \sigma_1 S_t \frac{\partial F}{\partial S} dW_1^{\mathbb{Q}} + \sigma_2 \frac{\partial F}{\partial \delta} dW_2^{\mathbb{Q}}. \end{aligned} \quad (13)$$

A futures contract requires no initial investment, so F must be a $\mathbb{Q}$ -martingale: its drift vanishes, $\mathbb{E}^{\mathbb{Q}}[dF] = 0$. Setting the bracket to zero gives the Gibson–Schwartz PDE,

$$\frac{\partial F}{\partial t} + (r - \delta) S \frac{\partial F}{\partial S} + \kappa(\alpha^{\mathbb{Q}} - \delta) \frac{\partial F}{\partial \delta} + \frac{1}{2} \sigma_1^2 S^2 \frac{\partial^2 F}{\partial S^2} + \rho \sigma_1 \sigma_2 S \frac{\partial^2 F}{\partial S \partial \delta} + \frac{1}{2} \sigma_2^2 \frac{\partial^2 F}{\partial \delta^2} = 0, \quad (14)$$

subject to the terminal condition $F(S_T, \delta_T, T) = S_T$.

4 Change of variables: time-to-maturity τ

In calendar time the terminal condition sits at $t = T$, which differs across contracts, undesirable for a network learning a single function valid across all maturities. Re-expressing in time-to-maturity $\tau = T - t$ collapses the terminal condition to the fixed point $\tau = 0$.

Let $\hat{F}(S, \delta, \tau) := F(S, \delta, t)$. Since $t = T - \tau$, the chain rule gives $\partial t / \partial \tau = -1$, so

$$\frac{\partial F}{\partial t} = -\frac{\partial \hat{F}}{\partial \tau}. \quad (15)$$

The spatial derivatives are unchanged, since S and δ depend on neither t nor τ . Substituting Equation 15 into Equation 14 and rearranging yields a forward equation in τ ,

$$\frac{\partial \hat{F}}{\partial \tau} = (r - \delta)S \frac{\partial \hat{F}}{\partial S} + \kappa(\alpha^{\mathbb{Q}} - \delta) \frac{\partial \hat{F}}{\partial \delta} + \frac{1}{2}\sigma_1^2 S^2 \frac{\partial^2 \hat{F}}{\partial S^2} + \rho\sigma_1\sigma_2 S \frac{\partial^2 \hat{F}}{\partial S \partial \delta} + \frac{1}{2}\sigma_2^2 \frac{\partial^2 \hat{F}}{\partial \delta^2}, \quad (16)$$

with the terminal condition becoming an initial condition,

$$\hat{F}(S, \delta, 0) = S. \quad (17)$$

This is a well-posed initial value problem and is the form used in the PINN: collocation points are sampled over (S, δ, τ) , with τ the interval between each observation date and the contract's delivery date.

5 Closed-form futures price

Equation 16 with $\hat{F}(S, \delta, 0) = S$ admits a closed-form, exponential-affine solution. Take the ansatz

$$\hat{F}(\tau, S, \delta) = S \exp[B(\tau)\delta + A(\tau)],$$

whose derivatives are

$$\begin{aligned} \frac{\partial \hat{F}}{\partial S} &= \frac{\hat{F}}{S}, & \frac{\partial^2 \hat{F}}{\partial S^2} &= 0, & \frac{\partial \hat{F}}{\partial \delta} &= B\hat{F}, & \frac{\partial^2 \hat{F}}{\partial \delta^2} &= B^2\hat{F}, \\ \frac{\partial^2 \hat{F}}{\partial S \partial \delta} &= B \frac{\hat{F}}{S}, & \frac{\partial \hat{F}}{\partial \tau} &= \hat{F}(B'\delta + A'). \end{aligned}$$

Substitution. Inserting these into Equation 16 and dividing by $\hat{F}$:

$$B'\delta + A' = (r - \delta) + \kappa(\alpha^{\mathbb{Q}} - \delta)B + \rho\sigma_1\sigma_2 B + \frac{1}{2}\sigma_2^2 B^2.$$

Matching the coefficient of δ and the constant term (the $\frac{1}{2}\sigma_1^2 S^2$ term drops out, since $\hat{F}_{SS} = 0$) gives two decoupled ODEs with $B(0) = A(0) = 0$:

$$B'(\tau) = -1 - \kappa B(\tau), \quad (18)$$

$$A'(\tau) = r + (\kappa\alpha^{\mathbb{Q}} + \rho\sigma_1\sigma_2)B(\tau) + \frac{1}{2}\sigma_2^2 B(\tau)^2. \quad (19)$$

Solving for $B(\tau)$. The linear ODE $B' + \kappa B = -1$ has integrating factor $e^{\kappa\tau}$, giving $e^{\kappa\tau}B = -\frac{1}{\kappa}e^{\kappa\tau} - C$. Applying $B(0) = 0$ fixes $C = -\frac{1}{\kappa}$, so

$$B(\tau) = -\frac{1 - e^{-\kappa\tau}}{\kappa}.$$

Solving for $A(\tau)$. Squaring,

$$B(\tau)^2 = \frac{1 - 2e^{-\kappa\tau} + e^{-2\kappa\tau}}{\kappa^2};$$

the $e^{-2\kappa\tau}$ term is the source of the second exponential in the result below. Substituting B and B^2 into A' and integrating term by term from 0 to τ (using $A(0) = 0$):

$$\begin{aligned} A(\tau) = & \left(r - \alpha^{\mathbb{Q}} + \frac{1}{2} \frac{\sigma_2^2}{\kappa^2} - \frac{\sigma_1 \sigma_2 \rho}{\kappa} \right) \tau + \frac{1}{4} \sigma_2^2 \frac{1 - e^{-2\kappa\tau}}{\kappa^3} \\ & + \left(\alpha^{\mathbb{Q}} \kappa + \sigma_1 \sigma_2 \rho - \frac{\sigma_2^2}{\kappa} \right) \frac{1 - e^{-\kappa\tau}}{\kappa^2}. \end{aligned}$$

Result.

$$\hat{F}(\tau, S, \delta) = S \exp[B(\tau)\delta + A(\tau)], \quad B(\tau) = -\frac{1 - e^{-\kappa\tau}}{\kappa},$$

with $A(\tau)$ as above. This matches the Schwartz (1997) Model 2 closed form term by term, under the identification $\hat{\alpha} = \alpha^{\mathbb{Q}}$ (his raw λ premium absorbed into $\alpha^{\mathbb{Q}} = \alpha^{\mathbb{P}} - \sigma_2 \lambda_2 / \kappa$, his T playing the role of τ).¹

6 Inequality constraints: the $\mathcal{L}_h$ terms

The data-misfit loss $\mathcal{L}_t$ and the PDE-residual loss $\mathcal{L}_f$ together pin the network output to the observed term structure and to the Gibson–Schwartz dynamics. Neither, however, constrains the inferred convenience yield $\hat{\delta}_\theta$ to lie in an economically admissible range: a path $\hat{\delta}_\theta$ may fit the observed futures and satisfy the pricing PDE while still taking values no rational holder of the physical commodity would tolerate. The inverse problem is in this sense ill-posed, and the inequality losses $\mathcal{L}_h$ act as a regularisation device that restores admissibility.

We group the constraints into two families. *Model-free* constraints follow from static no-arbitrage alone and hold independently of the Gibson–Schwartz specification; they bound the futures price $\hat{F}$, or the convenience yield $\hat{\delta}_\theta$, pointwise at a given maturity τ . *Model-implied* constraints (Section ??) instead restrict the shape of the futures curve across maturities and follow from the model’s term structure. Each constraint $g(\cdot) \geq 0$ is enforced through a one-sided hinge penalty,

$$\mathcal{L}_h = \frac{1}{N} \sum_{i=1}^N [\max(0, -g(x_i))]^2, \quad (20)$$

zero where the constraint holds and growing quadratically with the magnitude of any violation.

6.1 Model-free constraint I: cash-and-carry upper bound

The cash-and-carry strategy enforces an *upper* bound on the futures price. An arbitrageur who observes a futures price exceeding the cost of carrying the physical commodity to delivery can, at time t , borrow S_t at the risk-free rate, buy one unit of the commodity, pay

¹Schwartz (1997), eqs. (18)–(20), attributed to Jamshidian & Fein (1990) and Bjerksund (1991).

the storage and insurance cost of holding it, and simultaneously sell the futures. At delivery the commodity is handed over against the futures price, the loan is repaid, and the strategy locks in a riskless profit if

$$F_t > S_t e^{(r+u)\tau}, \quad (21)$$

where $u \geq 0$ is the per-unit, per-unit-time cost of physical storage (including insurance), expressed as a continuously compounded rate. No-arbitrage therefore requires $F_t \leq S_t e^{(r+u)\tau}$.

Crucially, the convenience yield does *not* enter the upper bound. The convenience yield is a benefit that accrues to the holder of physical inventory; it lowers the futures price relative to full carry and so can only relax the bound, never tighten it. A model-free arbitrage bound must not reference the latent state $\hat{\delta}_\theta$ being inferred, and this bound does not. The corresponding loss is

$$\mathcal{L}_{h,\text{cac}} = \frac{1}{N} \sum_i \left[\max(0, \hat{F}_t - S_t e^{(r+u)\tau}) \right]^2. \quad (22)$$

Taking logs of (21) and using the closed form $\hat{F}_t = S_t \exp[B(\tau)\hat{\delta}_\theta + A(\tau)]$ with $B(\tau) < 0$ shows the bound is equivalently a *lower* bound on the inferred convenience yield,

$$B(\tau)\hat{\delta}_\theta + A(\tau) \leq (r+u)\tau \iff \hat{\delta}_\theta \geq \frac{(r+u)\tau - A(\tau)}{B(\tau)}, \quad (23)$$

the inequality reversing because $B(\tau) < 0$. This is a useful internal consistency check: the cash-and-carry price ceiling and the economic-plausibility floor of Section 6.3 both act to bound $\hat{\delta}_\theta$ from below, so the two constraints reinforce rather than oppose one another.

6.2 Model-free constraint II: reverse cash-and-carry lower bound

The reverse strategy (short the physical commodity, invest the proceeds at r , and buy the futures) would appear to enforce a symmetric *lower* bound $F_t \geq S_t e^{r\tau}$. For commodities this symmetry fails, and the asymmetry is the defining economic feature of the market. To short the physical the arbitrageur must borrow the commodity from an inventory holder, and that holder forgoes the convenience yield for the duration of the loan; in a competitive market they demand compensation for it. The convenience yield is therefore a wedge that weakens, and in the limit removes, the lower bound. A futures price below $S_t e^{r\tau}$, a backwarddated curve, is not in general an arbitrage but the normal signature of a positive convenience yield.

The correct model-free statement carries the convenience-yield wedge explicitly. Writing $\delta_{\max}$ for the largest economically defensible convenience yield, the lower bound is

$$F_t \geq S_t e^{(r-\delta_{\max})\tau}, \quad (24)$$

with corresponding loss

$$\mathcal{L}_{h,\text{rcac}} = \frac{1}{N} \sum_i \left[\max(0, S_t e^{(r-\delta_{\max})\tau} - \hat{F}_t) \right]^2. \quad (25)$$

The naive bound $F_t \geq S_t e^{r\tau}$ is recovered only in the limit $\delta_{\max} \rightarrow 0$, i.e. for an asset with no convenience yield, where the commodity behaves like a financial forward. Because $\delta_{\max}$ is a calibrated plausibility ceiling rather than a quantity fixed by pure arbitrage, (25) sits at the boundary between the model-free and economic-plausibility families; we retain it here as the convenience-yield-corrected counterpart of the cash-and-carry bound.

6.3 Economic-plausibility constraint: the convenience-yield floor

The carry bounds of Sections 6.1–6.2 constrain the futures price. A separate constraint bounds the convenience yield directly from below. Unlike a volatility, the convenience yield may be negative: a commodity in deep contango, where storage cost dominates the benefit of holding inventory, carries $\delta_t < 0$. It cannot, however, fall without limit. Below some floor $\delta_{\min}$ a rational holder abandons the physical position rather than continue absorbing the net carrying cost, so the convenience yield observed in a market of willing holders is bounded below:

$$\hat{\delta}_\theta \geq \delta_{\min}, \quad \mathcal{L}_{h,\delta\text{-floor}} = \frac{1}{N} \sum_i \left[\max(0, \delta_{\min} - \hat{\delta}_\theta) \right]^2. \quad (26)$$

Both $\delta_{\min}$ and $\hat{\delta}_\theta$ are rates (dimension: time^{-1}), so the bound is dimensionally consistent and, being a property of the yield alone, carries no dependence on the price level S_t .

Setting $\delta_{\min}$. We fix $\delta_{\min}$ empirically rather than derive it. A decision-theoretic threshold, obtained by solving the holder’s optimal-stopping problem for the point of abandonment, was considered and rejected on grounds of internal consistency. Such a floor is a *structural* restriction, whereas δ_t here is a reduced-form Ornstein–Uhlenbeck factor whose Gaussian marginals place positive probability on every finite level. A hard structural floor would therefore contradict the very dynamics that the PDE residual $\mathcal{L}_f$ enforces, and would bias $\hat{\delta}_\theta$ upward in exactly the deep-contango regime the model is intended to capture. We instead take $\delta_{\min}$ as the empirical lower edge of the convenience-yield range in the Monte Carlo simulated data,

$$\delta_{\min} = -0.3 \text{ yr}^{-1}, \quad (27)$$

so that the constraint regularises the inversion without binding on economically plausible states. Written out with this value, the active penalty is

$$\mathcal{L}_{h,\delta\text{-floor}} = \frac{1}{N} \sum_i \left[\max(0, -0.3 - \hat{\delta}_\theta) \right]^2, \quad (28)$$

firing only where the inferred yield falls below -0.3 yr^{-1} and returning zero on all admissible states.

7 The composite objective

Training minimises a single scalar objective assembled from a data-misfit term, residual terms enforcing the model dynamics, and a family of no-arbitrage inequality penalties:

$$\hat{\mathcal{L}} := \underbrace{\lambda_t \hat{\mathcal{L}}_t}_{\text{Increasing Inputs}} + \underbrace{\lambda_b \hat{\mathcal{L}}_b + \lambda_f \hat{\mathcal{L}}_f}_{\text{PDE residuals}} + \underbrace{\sum_\alpha \lambda_{h_\alpha} \hat{\mathcal{L}}_{h_\alpha}}_{\text{No-arbitrage penalties}}. \quad (29)$$

The weights $\lambda_t, \lambda_b, \lambda_f, \{\lambda_{h_\alpha}\}$ are set adaptively by the WamOL loss-balancing mechanism, so that no single term dominates the gradient purely by scale. The inequality penalties $\hat{\mathcal{L}}_{h_\alpha}$ are derived in the companion note on no-arbitrage constraints; the two terms below are the data-misfit loss $\hat{\mathcal{L}}_t$ and the PDE-residual loss $\hat{\mathcal{L}}_f$.

8 Data-misfit loss $\hat{\mathcal{L}}_t$

The network's inferred convenience yield $\hat{\delta}_\theta$ produces a model futures price through the closed-form pricing map,

$$\hat{F}(S, \hat{\delta}_\theta, \tau) = S \exp[B(\tau)\hat{\delta}_\theta + A(\tau)], \quad (30)$$

with $B(\tau)$ and $A(\tau)$ as established in the pricing derivation. Taking logarithms linearises this in the latent yield,

$$\ln \hat{F} = \ln S + B(\tau)\hat{\delta}_\theta + A(\tau). \quad (31)$$

The data-misfit loss is the mean squared deviation of this model log-price from the observed log futures price $\ln F_i^{\text{obs}}$, taken over the N observations:

$$\hat{\mathcal{L}}_t = \frac{1}{N} \sum_{i=1}^N \left(\ln S_i + B(\tau_i)\hat{\delta}_\theta(t_i, S_i) + A(\tau_i) - \ln F_i^{\text{obs}} \right)^2. \quad (32)$$

Working in log-price rather than price has two consequences. The residual is *linear* in $\hat{\delta}_\theta$, which conditions the inversion well and matches the linear measurement equation used by the Kalman-filter baseline; and it is scale-invariant, so the loss behaves identically across price regimes rather than over-weighting periods of high absolute price.

9 PDE-residual loss $\hat{\mathcal{L}}_f$

The futures pricing PDE is recast as a residual by collecting every term on one side, defining the operator whose zero the true futures price satisfies:

$$r_{\text{PDE}} := \frac{\partial \hat{F}}{\partial \tau} - (r - \hat{\delta}_\theta)S \frac{\partial \hat{F}}{\partial S} - \kappa(\alpha^{\mathbb{Q}} - \hat{\delta}_\theta) \frac{\partial \hat{F}}{\partial \delta} - \frac{1}{2}\sigma_1^2 S^2 \frac{\partial^2 \hat{F}}{\partial S^2} - \rho\sigma_1\sigma_2 S \frac{\partial^2 \hat{F}}{\partial S \partial \delta} - \frac{1}{2}\sigma_2^2 \frac{\partial^2 \hat{F}}{\partial \delta^2}. \quad (33)$$

The partial derivatives of $\hat{F}$ are obtained by automatic differentiation of the network. Evaluating r_{PDE} at N_c sampled collocation points $(S, \delta, \tau)_j$ and penalising its squared magnitude gives the residual loss, first in compact form and then fully expanded:

$$\hat{\mathcal{L}}_f = \frac{1}{N_c} \sum_{j=1}^{N_c} \left(r_{\text{PDE}}^{(j)} \right)^2, \quad (34)$$

$$\hat{\mathcal{L}}_f = \frac{1}{N_c} \sum_{j=1}^{N_c} \left(\frac{\partial \hat{F}}{\partial \tau} - (r - \hat{\delta}_\theta)S \frac{\partial \hat{F}}{\partial S} - \kappa(\alpha^{\mathbb{Q}} - \hat{\delta}_\theta) \frac{\partial \hat{F}}{\partial \delta} - \frac{1}{2}\sigma_1^2 S^2 \frac{\partial^2 \hat{F}}{\partial S^2} - \rho\sigma_1\sigma_2 S \frac{\partial^2 \hat{F}}{\partial S \partial \delta} - \frac{1}{2}\sigma_2^2 \frac{\partial^2 \hat{F}}{\partial \delta^2} \right)_j^2. \quad (35)$$

A network whose outputs are consistent with the Gibson–Schwartz dynamics drives this residual to zero at every collocation point; deviations are penalised quadratically.

10 The term $\hat{\mathcal{L}}_b$

The term $\hat{\mathcal{L}}_b$ appears in the composite objective (29), grouped with the PDE residual, but its definition is not fixed here and is left to be specified.²

²Definition to be supplied.